

$$9) \ y = \frac{x^3 - x^2 - 3}{x^5 + 3}$$

$$10) \ y = \frac{x^4 + 6}{3 - 4x^{-4}}$$

$$11) \ y = \frac{4x^4 - 4x^2 + 5}{\frac{5}{2x^3 + 3}}$$

Critical thinking question:

- 12) A classmate claims that $\left(\frac{f}{g}\right)' = \frac{f'}{g'}$ for any functions f and g . Show an example that proves your classmate wrong.